

10th Grade Midterm Analysis of Decisions: C6–C7 Practice Activity 4

Contents

1 C6: Decision-Making Under Risk Using Probabilities	1
2 C7: Sensitivity Analysis of Probability-Based Decisions	3

Evaluated criteria

- C6** Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.
- C7** Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

1 C6: Decision-Making Under Risk Using Probabilities

Problems:

- 1.1. Community Theater Ticketing Strategy Under Demand Uncertainty
 - 1.2. Holiday Inventory Plan Selection Under Demand Uncertainty
 - 1.3. Routing System Choice Under Fuel and Congestion Uncertainty
-

1.1 Community Theater Ticketing Strategy Under Demand Uncertainty

A community theater must choose one of three ticketing strategies for the upcoming season: Strategy A, Strategy B, or Strategy C. Demand can be strong or weak with probabilities 0.60 and 0.40. The final payoff table (in thousands of dollars) is given below. Use Expected Value to select a strategy.

<i>Payoff table</i>				
State of nature	Probability	Strategy A	Strategy B	Strategy C
Strong demand	0.60	60	52	45
Weak demand	0.40	10	18	25

Questions/Tasks.

- C6** Use Expected Value to select a strategy.

C6 - Use Expected Value to select a strategy.

$$EV_A = 0.60(60) + 0.40(10) = 40, \quad EV_B = 0.60(52) + 0.40(18) = 38.4, \quad EV_C = 0.60(45) + 0.40(25) = 37.$$

Choose Strategy A.



1.2 Holiday Inventory Plan Selection Under Demand Uncertainty

A retail chain must select one of three inventory plans for the holiday period: Plan A, Plan B, or Plan C. Demand can be strong, moderate, or weak with probabilities 0.30, 0.50, and 0.20. The final payoff table (in thousands of dollars) is given below. Use Expected Value to select a plan.

<i>Payoff table</i>				
State of nature	Probability	Plan A	Plan B	Plan C
Strong demand	0.30	80	70	60
Moderate demand	0.50	50	55	45
Weak demand	0.20	10	30	40

Questions/Tasks.

C6 Use Expected Value to select a plan.

C6 - Use Expected Value to select a plan.

$$EV_A = 0.30(80) + 0.50(50) + 0.20(10) = 51, \quad EV_B = 0.30(70) + 0.50(55) + 0.20(30) = 54.5, \quad EV_C = 0.30(60) + 0.50(45) + 0.20(40) = 50.5$$

Choose Plan B.



1.3 Routing System Choice Under Fuel and Congestion Uncertainty

A shipping company must choose between two routing systems for the next season: System A or System B. Four states of nature summarize fuel and congestion conditions with probabilities 0.25, 0.30, 0.20, and 0.25. The final payoff table (in thousands of dollars) is given below. Use Expected Value to select a system.

<i>Payoff table</i>			
State of nature	Probability	System A	System B
State S_1	0.25	40	35
State S_2	0.30	30	28
State S_3	0.20	15	20
State S_4	0.25	-5	5

Questions/Tasks.

C6 Use Expected Value to select a system.

C6 - Use Expected Value to select a system.

$$EV_A = 0.25(40) + 0.30(30) + 0.20(15) + 0.25(-5) = 20.75, \quad EV_B = 0.25(35) + 0.30(28) + 0.20(20) + 0.25(5) = 22.4.$$

Choose System B.



2 C7: Sensitivity Analysis of Probability-Based Decisions

Problems:

2.1. Expected Value Comparison of Marketing Options

2.2. Expected Value Comparison of Production Plans

2.3. Expected Value Comparison of Zoning Options

2.1 Expected Value Comparison of Marketing Options

A publisher must choose between two marketing options, A and B, before knowing which market condition will occur. There are two possible states of nature: state S_1 , which occurs with probability p , and state S_2 , which occurs with probability $1 - p$. The objective is to select the option that yields the higher expected profit, measured in consistent monetary units. The payoff table below summarizes the profit associated with each option under each state of nature.

<i>Payoff table</i>		
	S_1 (p)	S_2 ($1 - p$)
A	24	12
B	18	16

Questions/Tasks.

C7 Determine the preferred alternative as a function of p , including the threshold/tie point.

C7 - Determine the preferred alternative as a function of p , including the threshold/tie point.

$$EV(A) = 12p + 12, \quad EV(B) = 2p + 16, \quad 10p - 4 > 0.$$

A if $p > 0.4$; B if $p < 0.4$; indifferent at $p = 0.4$.



2.2 Expected Value Comparison of Production Plans

A manufacturer must select one production plan, labeled A, B, or C, before knowing which market condition will occur. There are two possible states of nature: state S_1 , which occurs with probability p , and state S_2 , which occurs with probability $1 - p$. Each production plan generates a different profit depending on the realized state of nature, as summarized in the payoff table below. The objective is to determine which production plan maximizes expected profit as a function of p .

<i>Payoff table</i>		
	$S_1 (p)$	$S_2 (1 - p)$
A	32	4
B	20	14
C	12	10

Questions/Tasks.

C7 Determine the preferred alternative as a function of p , including the threshold/tie point.

C7 - Determine the preferred alternative as a function of p , including the threshold/tie point.

$$EV(A) = 28p + 4, \quad EV(B) = 6p + 14, \quad EV(C) = 2p + 10.$$

A if $p > \frac{5}{11}$; B if $p < \frac{5}{11}$; A and B tie at $p = \frac{5}{11}$; C is never optimal.



2.3 Expected Value Comparison of Zoning Options

A city council must choose between two zoning options, labeled A and B, before knowing which future condition will occur. There are two possible states of nature: state S_1 , which occurs with probability p , and state S_2 , which occurs with probability $1 - p$. Each zoning option generates a different net return depending on the realized state of nature, as shown in the payoff table below. The objective is to determine which zoning option yields the higher expected return as a function of p .

<i>Payoff table</i>		
	$S_1 (p)$	$S_2 (1 - p)$
A	26	6
B	20	12

Questions/Tasks.

C7 Determine the preferred alternative as a function of p , including the threshold/tie point.

C7 - Determine the preferred alternative as a function of p , including the threshold/tie point.

$$EV(A) = 20p + 6, \quad EV(B) = 8p + 12, \quad 12p - 6 > 0.$$

A if $p > 0.5$; B if $p < 0.5$; indifferent at $p = 0.5$.

